

Samya Aggarwal

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EDUCATION

Thapar Institute of Engineering and Technology
BE in Computer Science and Business Systems
CGPA: 9.74/10.0

Patiala, Punjab
Sep 2023 – May 2027

Cambridge International School (CBSE)
Class XII (CBSE): 95.16%

Amritsar, Punjab
May 2022 – May 2023

SKILLS

Languages: Python, C++, C, JavaScript, SQL
Web: Django, Django REST Framework, Node.js, FastAPI, Django Channels, WebSockets
Python Libraries: Pandas, NumPy, Matplotlib, SHAP
Tools/Cloud: Git/GitHub, Redis, AWS, Linux
Databases: MySQL, PostgreSQL, SQLite
Concepts: Concurrency, Multithreading, Distributed systems, Agile Development, CI/CD, REST APIs
Coursework: Data Structures and Algorithms (DSA), Database Management Systems (DBMS), Object Oriented Programming (OOP), Operating Systems (OS), Computer Networks

PROFESSIONAL EXPERIENCE

Zylectra **Venture Lab Thapar**
Software Development Intern *Sep 2025 – Oct 2025*

- Built scalable Node.js backend services handling concurrent booking requests with robust conflict-resolution logic.
- Resolved API failures and performance bottlenecks using logging, profiling and structured debugging approaches.
- Integrated ML-based recommendations via FastAPI and improved API response latency by ~25% by introducing Redis caching.

InveshoAI **Remote (US-Based)**
Software Engineering Intern *Jun 2025 – Aug 2025*

- Built a FastAPI-based automation service to streamline internal workflows, reducing repetitive manual operations.
- Engineered a lightweight comment-generation pipeline using LLM APIs to support automated engagement workflows.
- Built and deployed tooling supporting a Product Hunt launch, contributing to user acquisition across 30+ countries.

PROJECTS

SyncRoomies: Roommate Selection Platform [GitHub](#)

- Designed and deployed a roommate matching platform serving 30+ users with real-time chat and profile matching.
- Developed real-time messaging using Django Channels & WebSockets for concurrent interactions in a distributed system.
- Implemented Selenium-based automated tests with CI/CD, improving reliability and reducing production bugs.

SafeSpace: AI-Powered Mental Health Platform [GitHub](#)

- Designed a scalable and maintainable FastAPI backend for real-time multimodal inference with 85% model accuracy.
- Engineered a custom ESP32-based wearable with 4 biosensors (ECG, EDA, EMG, Temp) for continuous data streaming.
- Deployed the solution on Vercel with a FastAPI backend, adding a SHAP-based XAI layer for transparency.

SmartKheti: AI-Powered Crop Disease Detection [GitHub](#)

- Processed and analyzed image datasets to train CNN and Autoencoder models for crop disease detection (80% accuracy).
- Built a full-stack web app with Django backend and responsive JavaScript frontend for image uploads.
- Added a crop recommendation module based on climate and regional data for improved decision-making.

ACHIEVEMENTS & LEADERSHIP

- Amazon ML Summer School 2025:** Selected among **3,000 students** nationwide for an intensive program in Deep Learning, Generative AI, and Reinforcement Learning.
- Reliance Foundation Scholar:** Selected as one of 5,000 students nationwide; earned a **Rs. 2 lakh** merit scholarship.
- Hackathons:** Awarded **3rd Prize** at the **Indian-Israeli Hackathon**; **selected for HackPrinceton'26**.
- Academic Excellence:** Secured **3rd rank** in first and second year (2023–25); awarded a **50% tuition fee** scholarship.
- LeetCode:** Solved 250+ DSA problems across Graphs, DP, Trees, Greedy and Advanced Data Structures.
- Hacktoberfest 2025 Supercontributor:** Shipped 6+ PRs improving features, bug fixes, and documentation.
- Creative Computing Society (Executive Member):** Organized technical workshops and contributed to projects.